

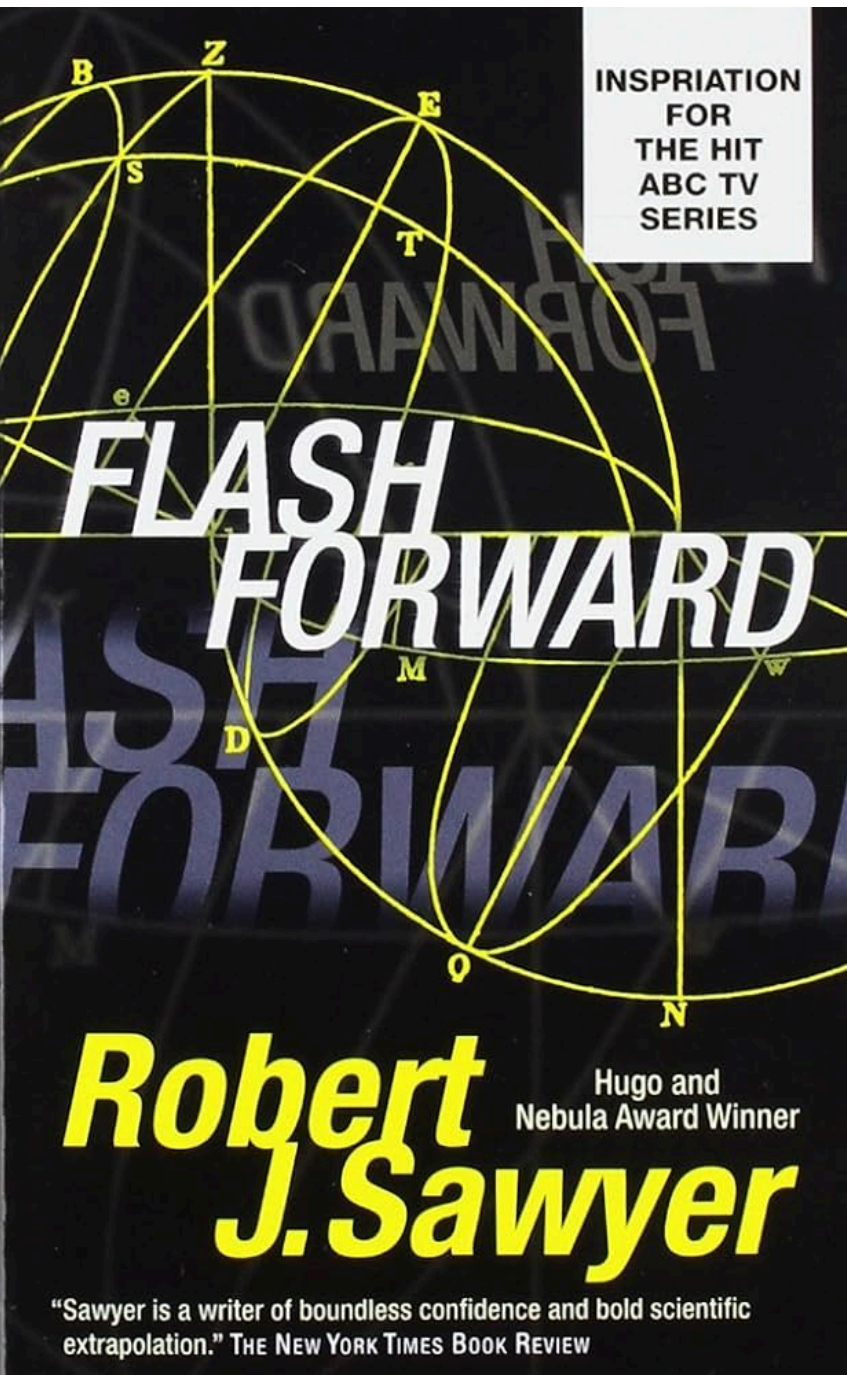


***Indico:** the 20 year history and evolution of an open-source project at **CERN***

Dominic Hollis & Tomas Roun - Indico Team (CERN)

# What is CERN?







## Largest particle physics lab in the world

- ~6.2 Square kilometers
- ~700 buildings on multiple sites
- 2500+ members of staff
- 12000+ visiting scientists
- 150'000+ visitors each year





**What exactly does CERN do?**

**CERN**

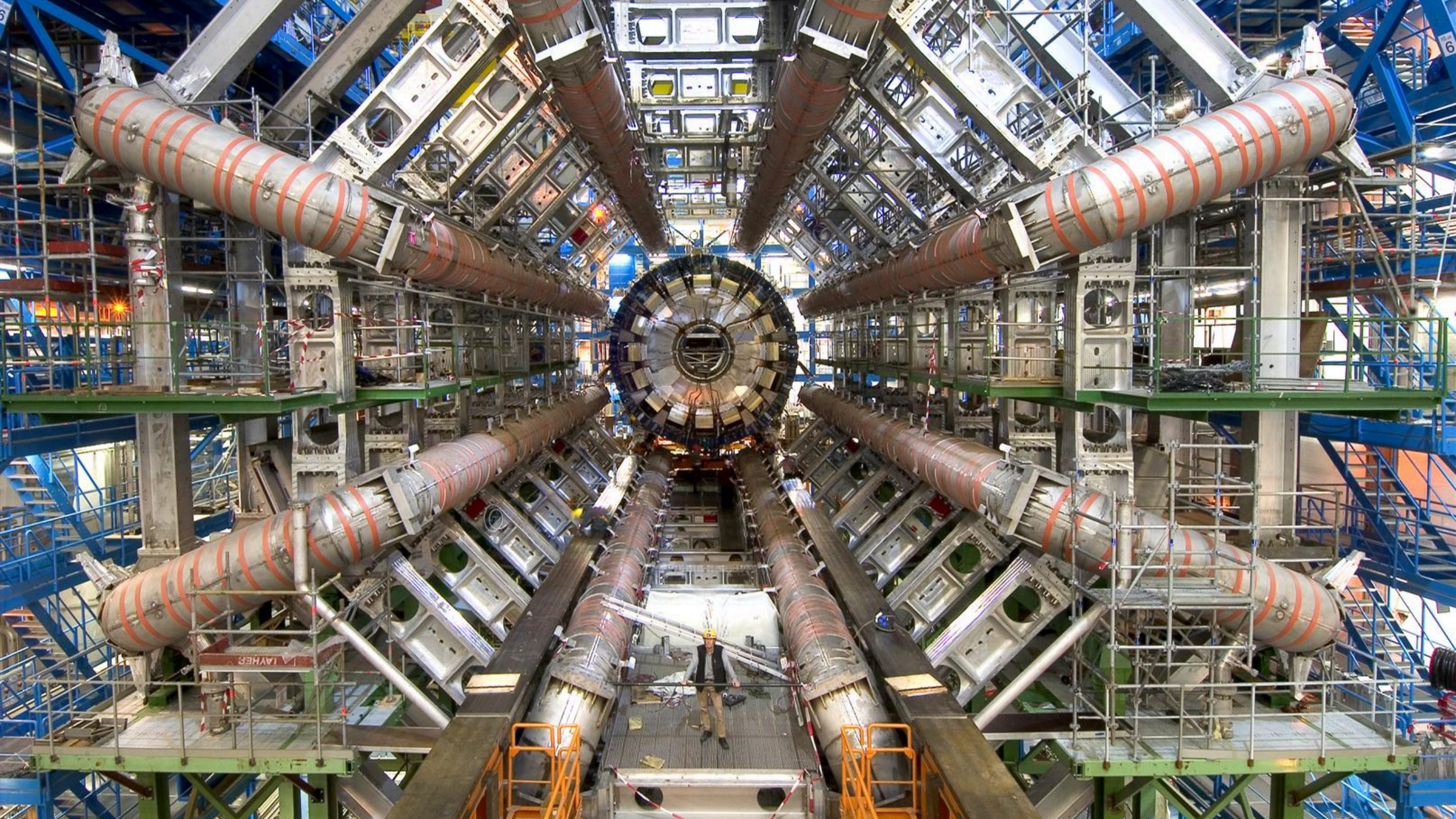
**PARTICLE**

**PARTICLE**

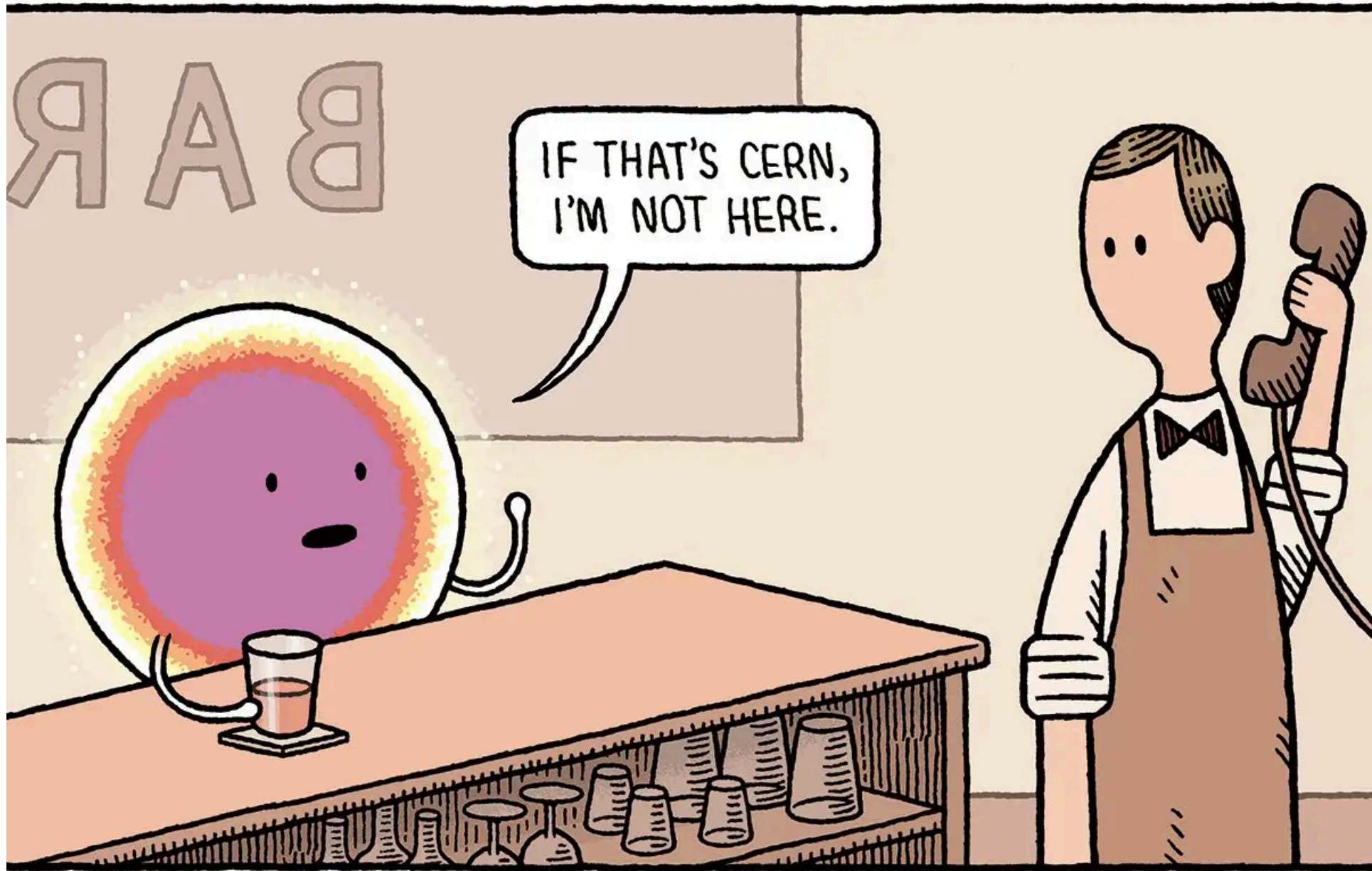












TOM GAULD for NEW SCIENTIST



# WHERE THE WEB WAS BORN

In the offices of this corridor, all the fundamental technologies of the World Wide Web were developed.

Started in 1990 from a proposal made by Tim Berners-Lee in 1989, the effort was first divided between an office in building 31 of the Computing and Networking Division (CN) and one in building 2 of the Electronics and Computing for Physics Division (ECP).

In 1991 the team came together in these offices, then belonging to ECP. It was composed of two CERN staff members, Tim Berners-Lee (GB) and Robert Cailliau (BE), aided by a number of Fellows, Technical Students, a Coopérant and Summer Students.

At the end of 1994 Tim Berners-Lee left CERN to direct the WWW Consortium (W3C), a world-wide organization devoted to leading the Web to its full potential. The W3C was founded with the help of CERN, the European Commission, the Massachusetts Institute of Technology (MIT), the Institut National pour la Recherche en Informatique et en Automatique (INRIA), and the Advanced Research Projects Agency (ARPA).

In 1995 Tim Berners-Lee and Robert Cailliau received the ACM Software System Award for the World Wide Web. In 2004, Tim Berners-Lee was awarded the first Millennium Technology Prize by the Finnish Technology Award Foundation.

The CERN Library  
June 2004



# CERN ❤️ Open Source





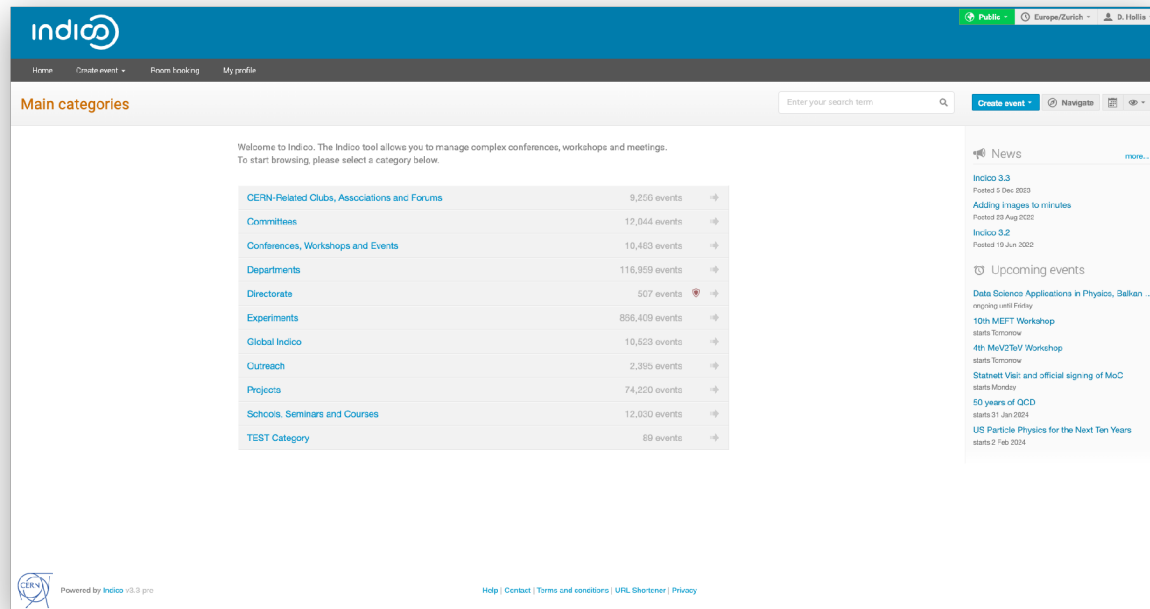
# CERN ❤️ Open Source

👉 <https://opensource.cern/>

👉 <https://github.com/CERN/awesome-cern>

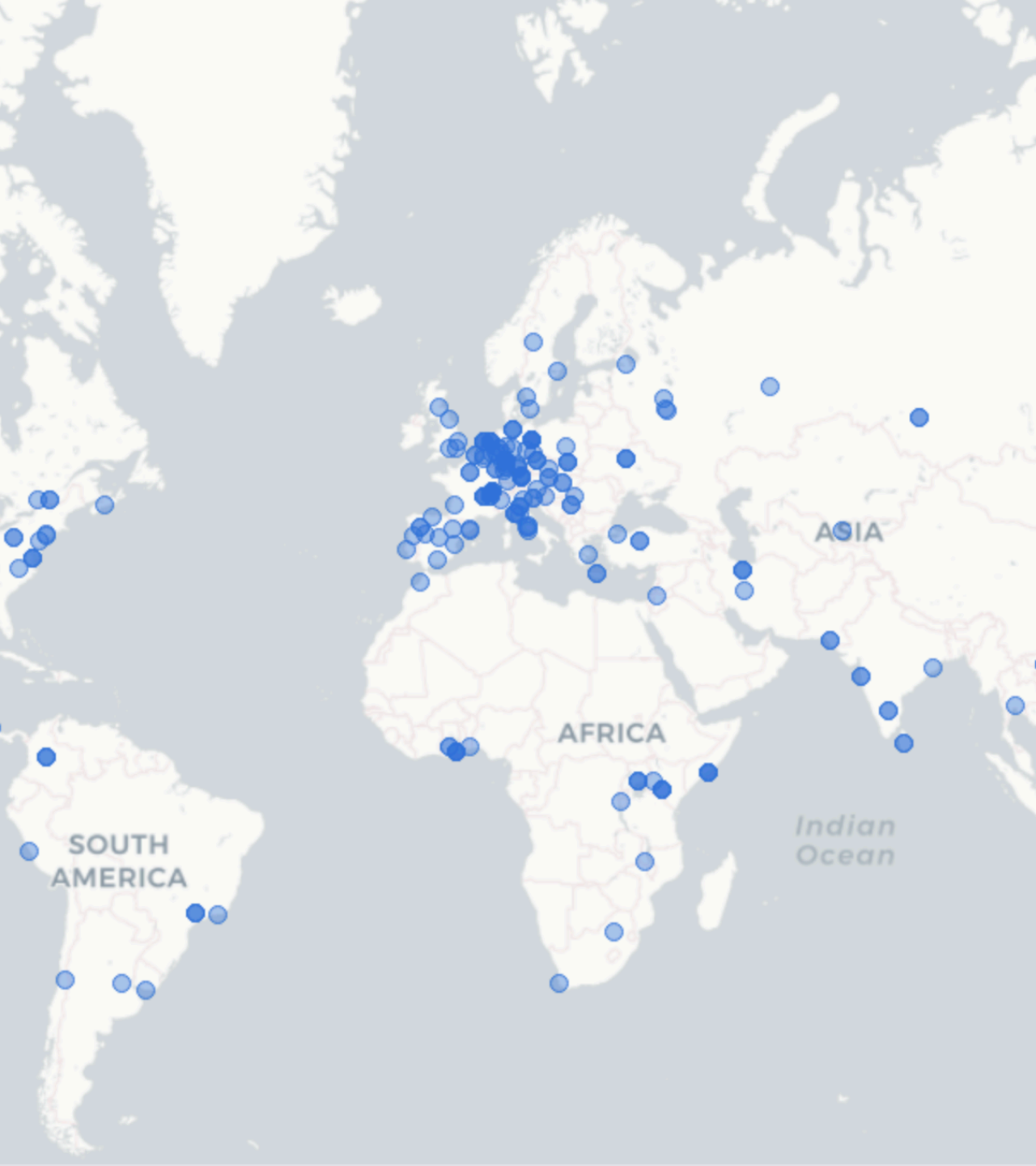


# **What is Indico?**



- **Event Management System**
- **Collaborative effort - MIT License**
- **Core Developed at CERN**
- **With contributions from the United Nations, Max-Planck Institute for Physics and many others!**
- **70+ developers** over the years





***The most popular event  
management system you never  
heard about***

- **300+ servers**
- **> 350K users**
- **20+ years** of development
- Initial growth in research, but growing beyond it
  - [indico.un.org](https://indico.un.org)
  - [events.canonical.org](https://events.canonical.org)
  - [indico.gnome.org](https://indico.gnome.org)
  - [lpc.events](https://lpc.events)

# Under the hood



We ♥ Python

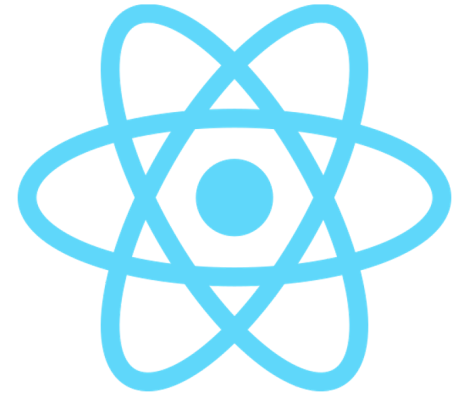
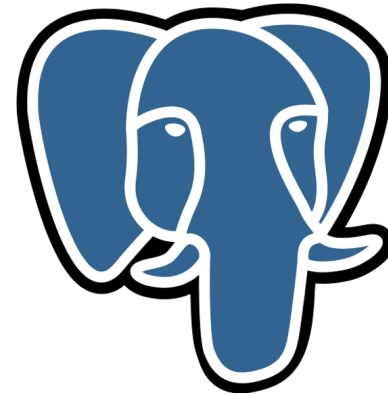


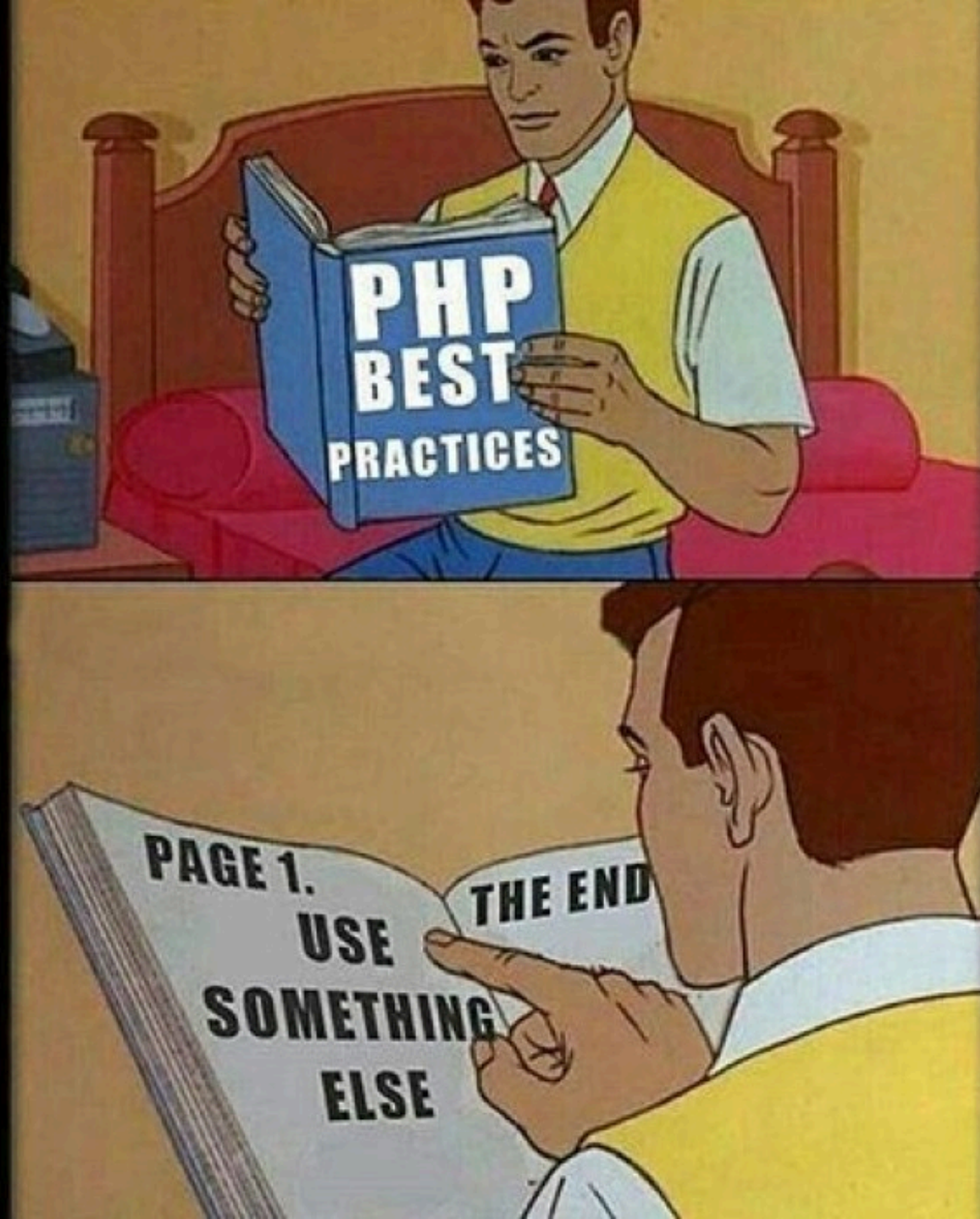
**Flask**

SQLAlchemy

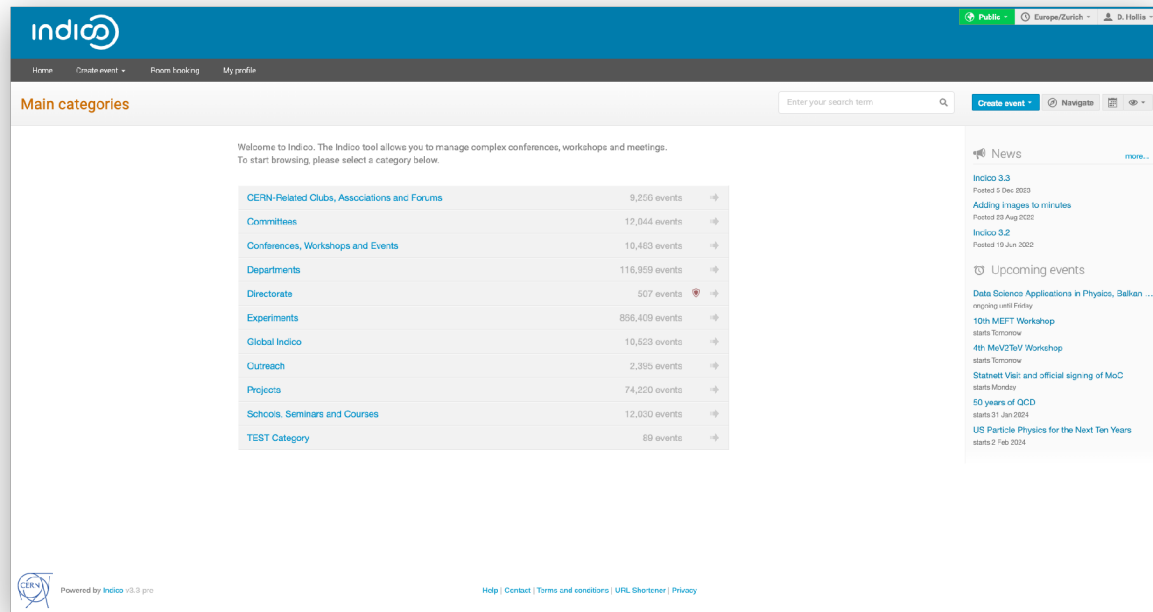


python™









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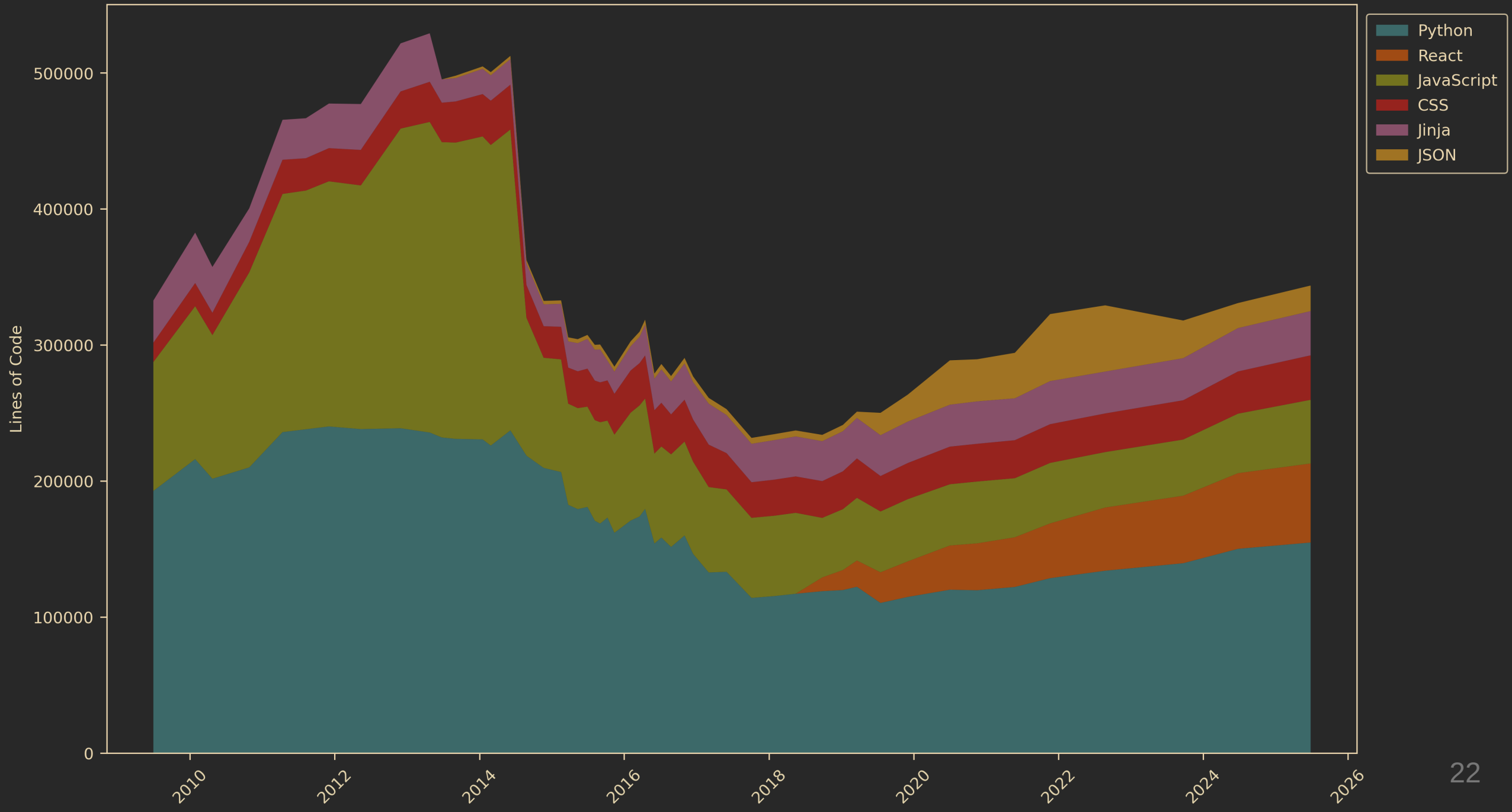
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  - [lpc.events](https://lpc.events)



# **WIP: The Story Hidden in the Code**

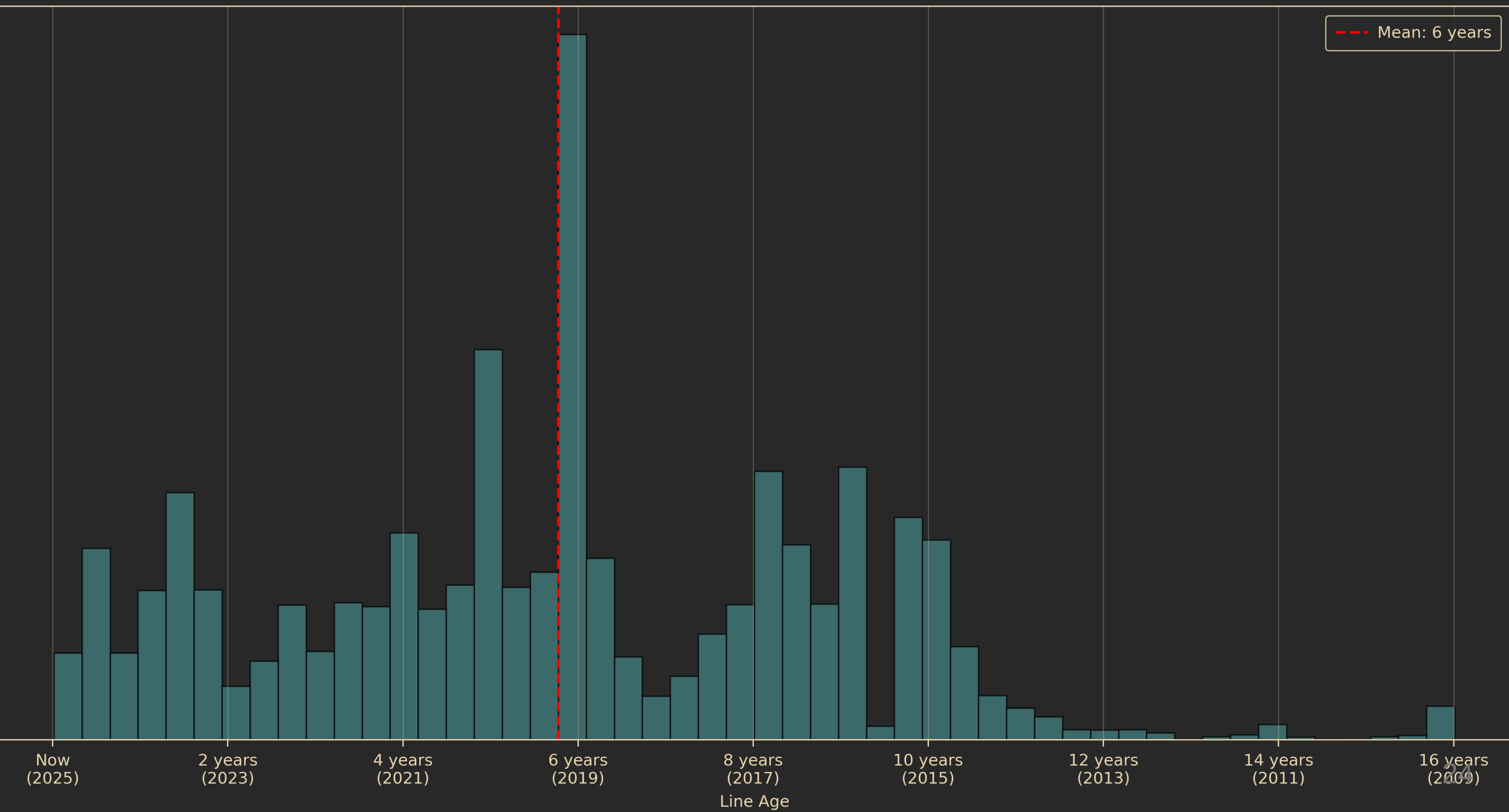
Language Composition Over Time



# Dealing with Technical Debt

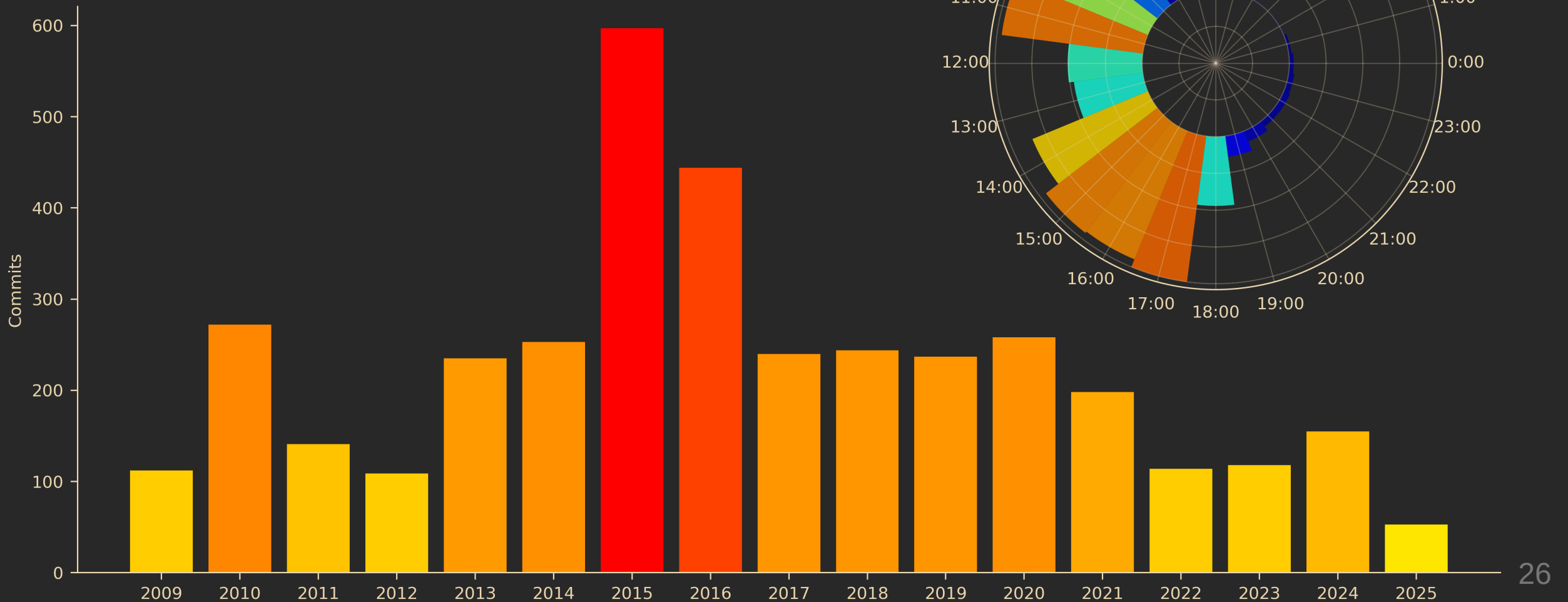


Line Age Distribution



# **Maintaining a healthy work-life balance**

After-hours Commits (commits made outside regular working hours)

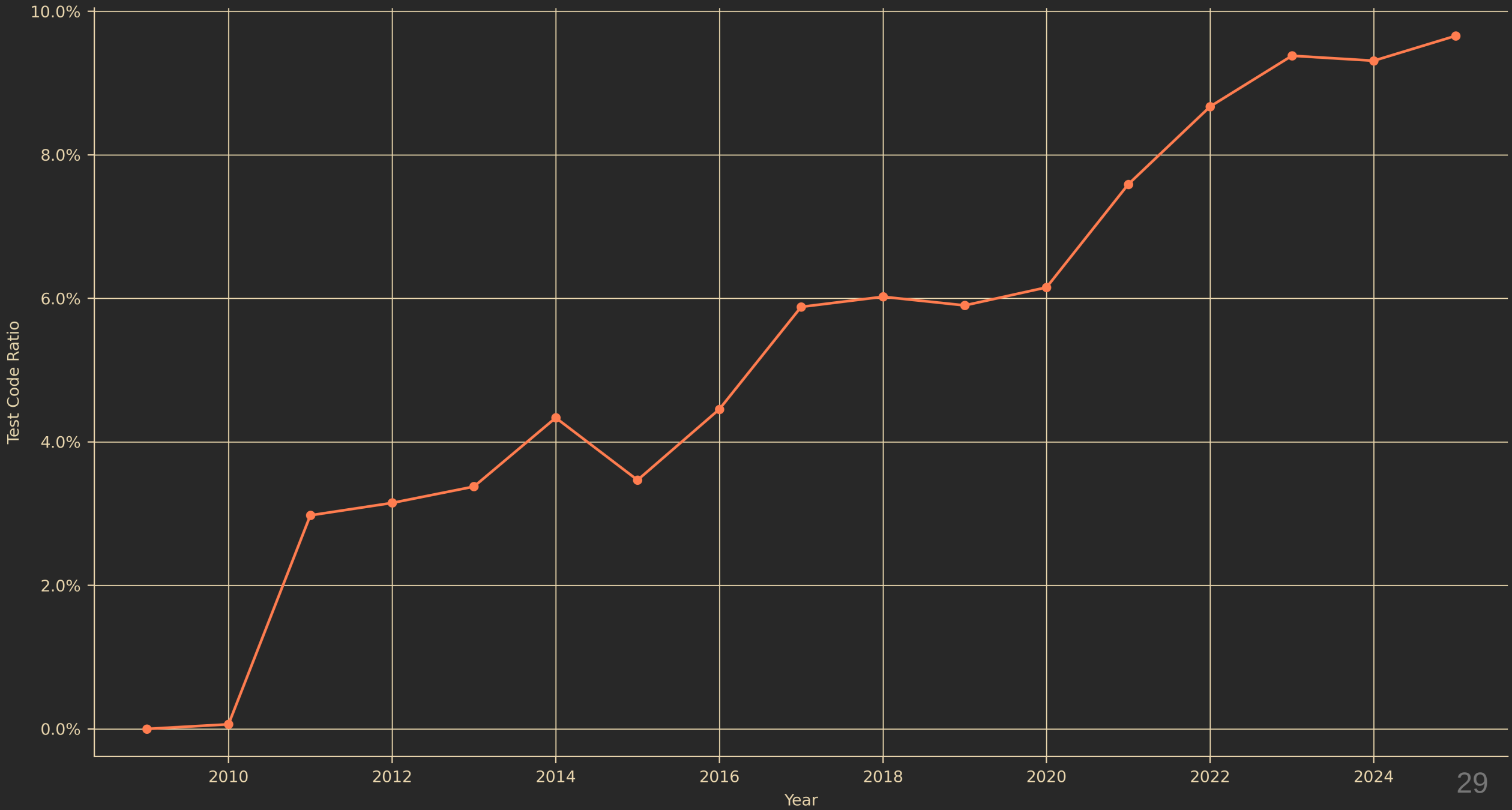




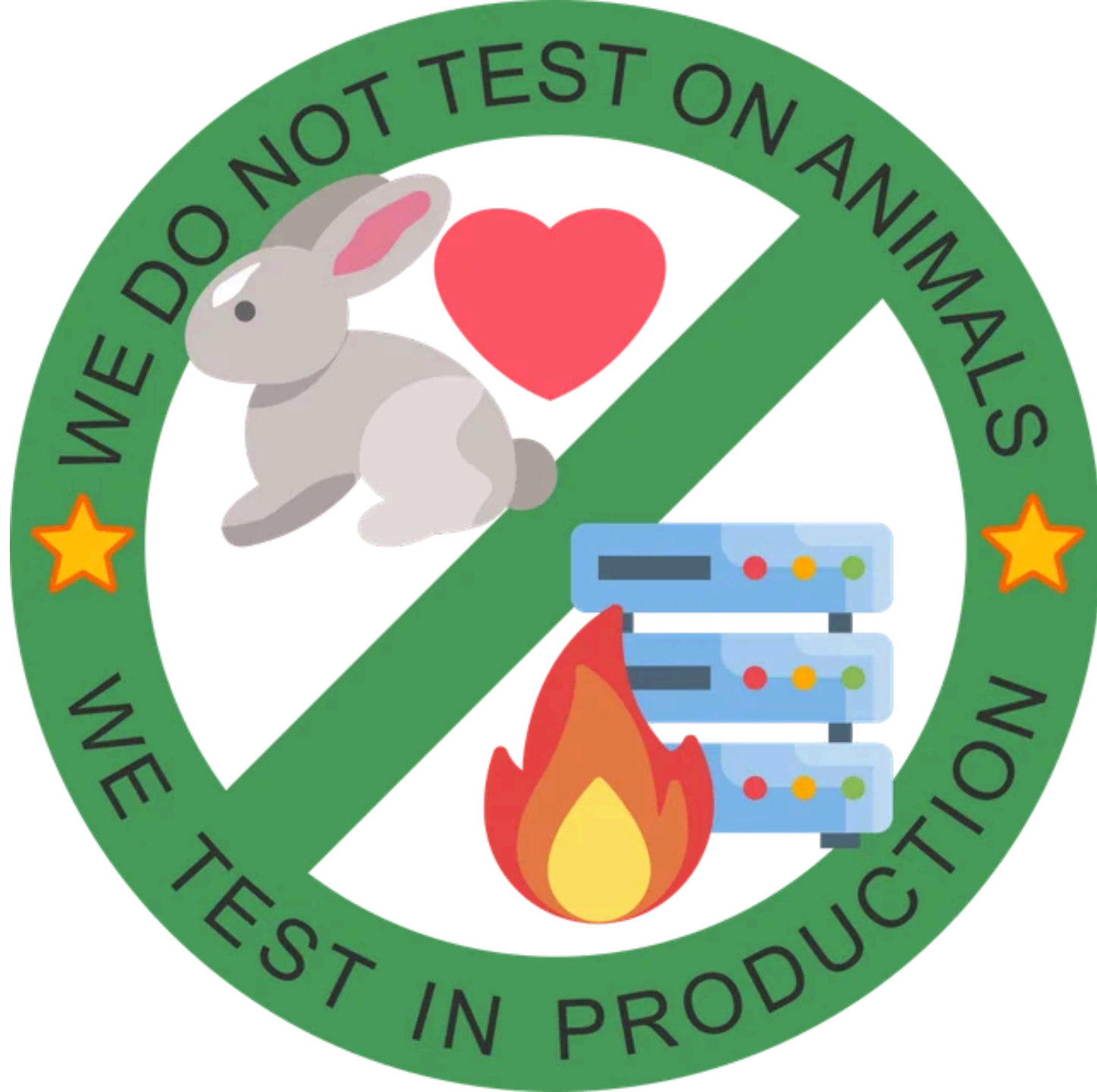


# Tests

Test Code Ratio Over Time

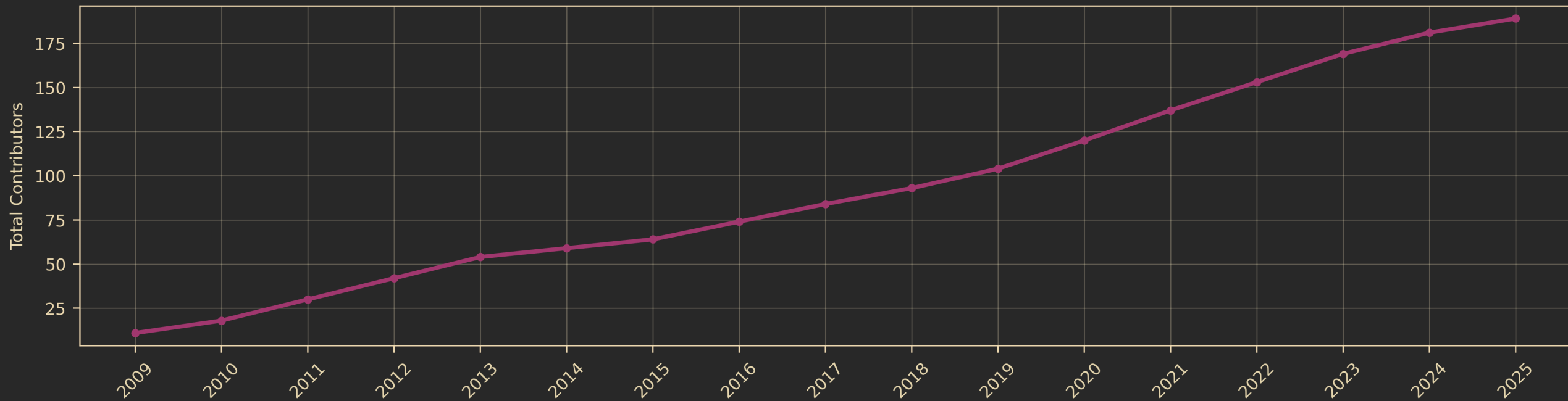




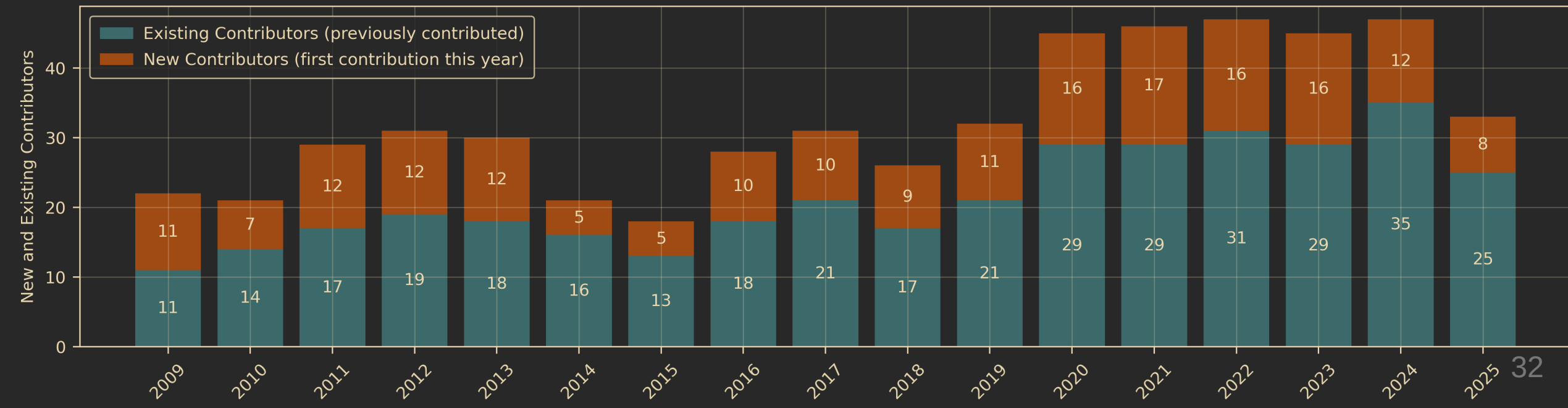


# Contributors

All Contributors (Cumulative)



New Contributors



# Translations

 Welcome to Indico  
 Willkommen bei Indico  
 Bienvenidos a Indico  
 Bienvenue dans Indico  
 Benvenuti in Indico  
 Üdvözljük az Indico-ban  
 Witaj w Indico  
 Bem vindo ao Indico

 Välkommen till Indico  
 Indico'ya hoşgeldiniz  
 Vítejte v Indicu  
 Индикод тавтай морил  
 Ласкаво просимо до Indico  
 Indico  
 Indico





# WIP: To close it off..

- Indico has been through many changes over the last (more than) two decades
- We've switched languages, technologies and databases countless times
- Some changes were more drastic than others
- what makes this really worth it is seeing the impact of our work
- Do not blindly accept everything people ask for, especially if the maintenance burden is large.
- we're looking forward to the next 20 years
- we have some merch!

 [getindico.io](https://getindico.io)

 [@getindico](https://matrix.org/join/getindico)

 [@getindico](https://twitter.com/getindico)

 [#indico:matrix.org](https://matrix.org/join/getindico)

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